EDUMIND: ACADEMIC KNOWLEDGE ASSISTANT

NLP Mini-Project Specification & Empirical Evaluation Report
Fine-Tuned Transformers • Semantic Retrieval • RAG • WordNet Analysis

1. Executive Summary & Abstract

EduMind is an intelligent multilingual academic knowledge assistant designed to solve the lexical gap between formal institutional circulars and natural student queries. The system incorporates all five mandatory NLP preprocessing stages, domain-adapted BERT intent classification (achieving 98.76% accuracy), FAISS dense semantic search (0.8750 MRR), anti-hallucination grounded RAG generation, text summarization, and Indian language routing (Hindi/Marathi).

2. Mandatory Classical NLP Preprocessing

- **Sentence Boundary Identification:** NLTK Punkt model segmenting raw documents into sentence units.
- **Tokenization & Lexical Stats:** Word-level tokenization computing lexical diversity (Type-Token Ratio).
- **Lemmatization vs Stemming:** WordNetLemmatizer with POS tag mapping vs heuristic Porter Stemmer.
- **Syntactic Analysis (POS Tagging):** Penn Treebank tag assignment (NN, VB, MD, JJ, WRB).
- **Semantic Analysis (WordNet):** Synset hierarchies, hypernyms, and Path/Wu-Palmer similarity metrics.

3. Model Comparison Benchmark (Section 26)

Model Architecture	Type	Accuracy	Weighted F1	Macro F1
TF-IDF + Logistic Reg.	Classical ML	98.35%	98.39%	98.39%
Pretrained BERT (Untrained)	Zero-Shot	8.50%	8.08%	8.08%
Fine-Tuned BERT (EduMind)	Fine-Tuned Transformer	98.76%	98.76%	98.65%

4. Information Retrieval Evaluation (Section 27)

Retrieval Method	Precision@1	Precision@3	Recall@3	MRR	Latency
Dense Semantic (FAISS + MiniLM)	83.33%	72.22%	83.33%	0.8750	45.19 ms
Sparse Keyword (TF-IDF)	91.67%	83.33%	91.67%	0.9583	0.23 ms
Hybrid RRF (Dense + Keyword)	91.67%	86.11%	100.0%	0.9583	46.10 ms

5. Generation Quality (ROUGE & BERTScore)

- **ROUGE-1 F1:** 0.2775 | **ROUGE-2 F1:** 0.2003 | **ROUGE-L F1:** 0.2545
- **BERTScore (Contextual Token Embeddings):** Precision: 0.4598 | Recall: 0.7705 | F1: 0.5754

Note on BERTScore: Per Section 29 of the specification, BERTScore is applied specifically to evaluate grounded answers and summaries against ground truth reference text, capturing contextual semantic meaning rather than mere n-gram overlap.

6. Key Viva Voce Concepts

Q1: Why fine-tune BERT for intent classification?
Fine-tuning adapts the transformer self-attention representations specifically to institutional vocabulary, providing deterministic intent classification with low latency and zero external API dependencies.

Q2: What is the benefit of RAG over vanilla LLM prompting?
RAG grounds the answer strictly in retrieved official passages with exact page citations, eliminating hallucinated policies.